PCM · SCENARIO TEST · 12 KINDS OF REQUEST · 3,464 REQUESTS

Which method helps, and where

We ran each of our four methods on 12 kinds of real requests — short questions, math, code, documents, chats, Persian — and graded every answer against the known correct answer. This is the short answer.

TURN ON

Reorder for cache

Puts the fixed gateway policy before the user's text.

46–99%

of each prompt can come from the cache, instead of 0%. Answers got slightly better in 5 of 12 kinds of request and worse in 0.

KEEP OFF

Summarize old chat

Replaces old chat turns with a short summary.

−43.3 vs **+10.1**

Destroys answers when the question refers back; helps when the user changed topic. The gateway can't tell which one it is facing.

KEEP OFF

Rewrite the question

A model rewrites the user's question to be clearer.

0 of 12 better

even with a strong rewriting model (GPT-5.4-Mini); 2 got worse. Our small model was worse still — in math it changed the facts of the problem.

ONLY LONG DOCUMENTS

Remove unimportant words

Deletes words a model thinks don't matter.

−54% tokens

on long documents, for −4.3 points. Its 5,000-token rule is essential: without it, scores fell 24 to 51 points everywhere.

How to read the numbers. Every score is out of 100. A change is in **points**: +5 means 5 more out of 100 answered right. Asking the model the *same* question twice already moves a score by up to ±6 points by chance, so we only call a change real when it is clearly bigger than that.

1 · HOW WE TESTED

Same question, with and without the method

1 Take a real question that has a known correct answer → **2** Send it twice: as it is, and after the method changed it → **3** GPT-5.4-Mini answers both versions → **4** Grade both answers against the correct one, and compare

No AI judge: each dataset has its own answer key (the right letter, the right number, code tests that pass, or overlap with the correct words). Every request also carries the same ~1,100-token gateway policy, as it would in production.

	KIND OF REQUEST — WHAT IT LOOKS LIKE	DATASET	QUESTIONS	SCORE WITHOUT METHODS
1	**Short factual question** “who wrote the poem for whom the bell tolls”	NQ-Open	300	49
2	**Multiple-choice exam question** An exam question with options A–D (medicine, law, physics…)	MMLU	300	81
3	**Math word problem** “Becky ate 3 more slices than Jake… how many slices in total?”	GSM8K	300	94
4	**Write code (code inside the prompt)** “Finish this Python function” — graded by running its tests	HumanEval	164	93
5	**Instructions with strict format rules** “Write an outline, highlight 15 sections, under 100 words…”	IFEval	300	86
6	**Question about one short passage** A Wikipedia paragraph, then a question about it	SQuAD	300	89
7	**Question over 10 retrieved documents** 10 retrieved articles (2 useful), question needs both	HotpotQA	300	74
8	**Question about a long document (5k+ tokens)** A 5,000–8,000-token story, then a multiple-choice question	QuALITY	300	78
9	**Chat: last question refers back** Long chat about a story; last question: “Where did she go?”	CoQA	300	76
10	**Chat: user switches to a new topic** Long chat about a story; last question is about something else	CoQA + NQ-Open	300	33
11	**Persian multiple-choice question** The same kind of exam questions, in Persian	Global-MMLU (fa)	300	69
12	**Persian question about a passage** A Persian Wikipedia paragraph, then a question	PersianQA	300	76

Two settings per method. *Shipped* = with its safety rules (e.g. “skip code”, “only compress 5,000+ tokens”). *Safety rules off* = what the method does to that content if nothing stops it.

“Not used” means the shipped safety rules correctly left that kind of request alone — the answer is exactly the same as without the method.

Every method, every kind of request

Shipped settings. Numbers are the change in score, in points.

better · worse · no real change (within chance) · not used — skipped by its rules

	Reorder for cache	Summarize old chat	Rewrite (small model)	Remove unimportant words
Short fact question	+5.7 better	not used	−2.3 no real change	not used
Multiple choice	+3.3 no real change	not used	−1.0 no real change	not used
Math problem	+0.7 no real change	not used	−2.0 worse	not used
Write code	−1.2 no real change	not used	not used	not used
Strict format rules	+0.0 no real change	not used	−1.3 worse	not used
One passage	+2.9 better	not used	+0.2 no real change	not used
10 documents	+4.6 better	not used	+1.4 better	not used
Long document (5k+)	+2.7 no real change	not used	+1.0 no real change	−4.3 no real change
Chat: refers back	+6.0 better	−43.3 worse	−0.1 no real change	not used
Chat: new topic	−0.6 no real change	+10.1 better	−0.4 no real change	not used
Persian multiple choice	+2.7 no real change	not used	−3.7 worse	not used
Persian passage	+3.4 better	not used	−0.5 no real change	not used

Reorder is the only column with no red at all.

Summarize has one big red and one clear blue — opposite results on the two chats.

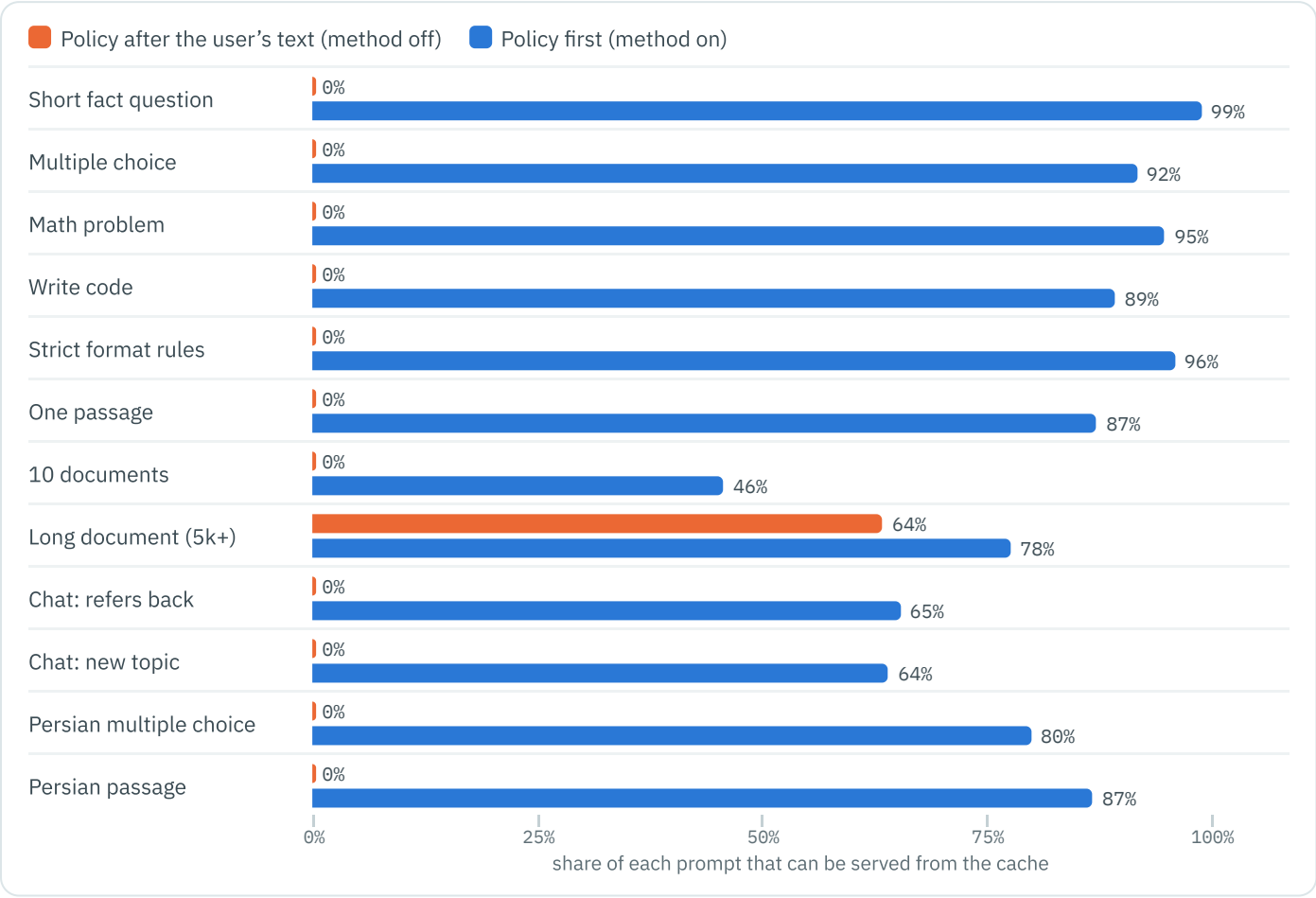
Rewrite has several small reds and one small blue — a strong rewriting model turned none of it blue (page 5).

Remove words only acts on long documents — there it is borderline.

3 · METHOD 1 · REORDER FOR CACHE

Policy first: more cache, and no harm

AI providers reuse work for the part of a prompt that is identical to an earlier one — but only from the very first word onward. Our policy is the same on every request, so putting it **first** lets that whole part come from the cache. Put after the user's text, it never matches.



Why some bars are shorter. In "10 documents" and "long document" most of each prompt is the document itself, which is new on every request, so the fixed part is a smaller share. Long documents are partly cached even when the policy is last, because several questions ask about the same story.

Quality. Answers were better in 5 kinds of request (up to +6.0 points) and worse in none. Part of that gain may come from the answer instructions sitting before the question rather than after it — either way, the new order does not hurt.

4 · METHOD 2 · SUMMARIZE OLD CHAT

It depends on the chat — and we can’t know in advance

Last question refers back

76 → 32

−43.3 points · 145 of 300 answers lost

User changed topic

33 → 43

+10.1 points · 36 answers gained

Both cut the chat’s tokens by the same 68%.

Refers back: the summary dropped the detail the question needed

ORIGINAL QUESTION `Does he consider himself a hero?`

ANSWER WITHOUT METHOD	ANSWER WITH METHOD	CORRECT ANSWER
No.	unknown	No.

The summary kept the story (a man threw his shoes at a president) but not what he said about himself.

Refers back: the summary made the model pick the wrong person

ORIGINAL QUESTION `Who caught her rapturously?`

ANSWER WITHOUT METHOD	ANSWER WITH METHOD	CORRECT ANSWER
Patsy	Mr. Watson	Patsy

Wrong with confidence is worse than “unknown”: nobody notices.

New topic: removing the unrelated chat helped

ORIGINAL QUESTION `who won the mens figure skating in 2018 olympics`

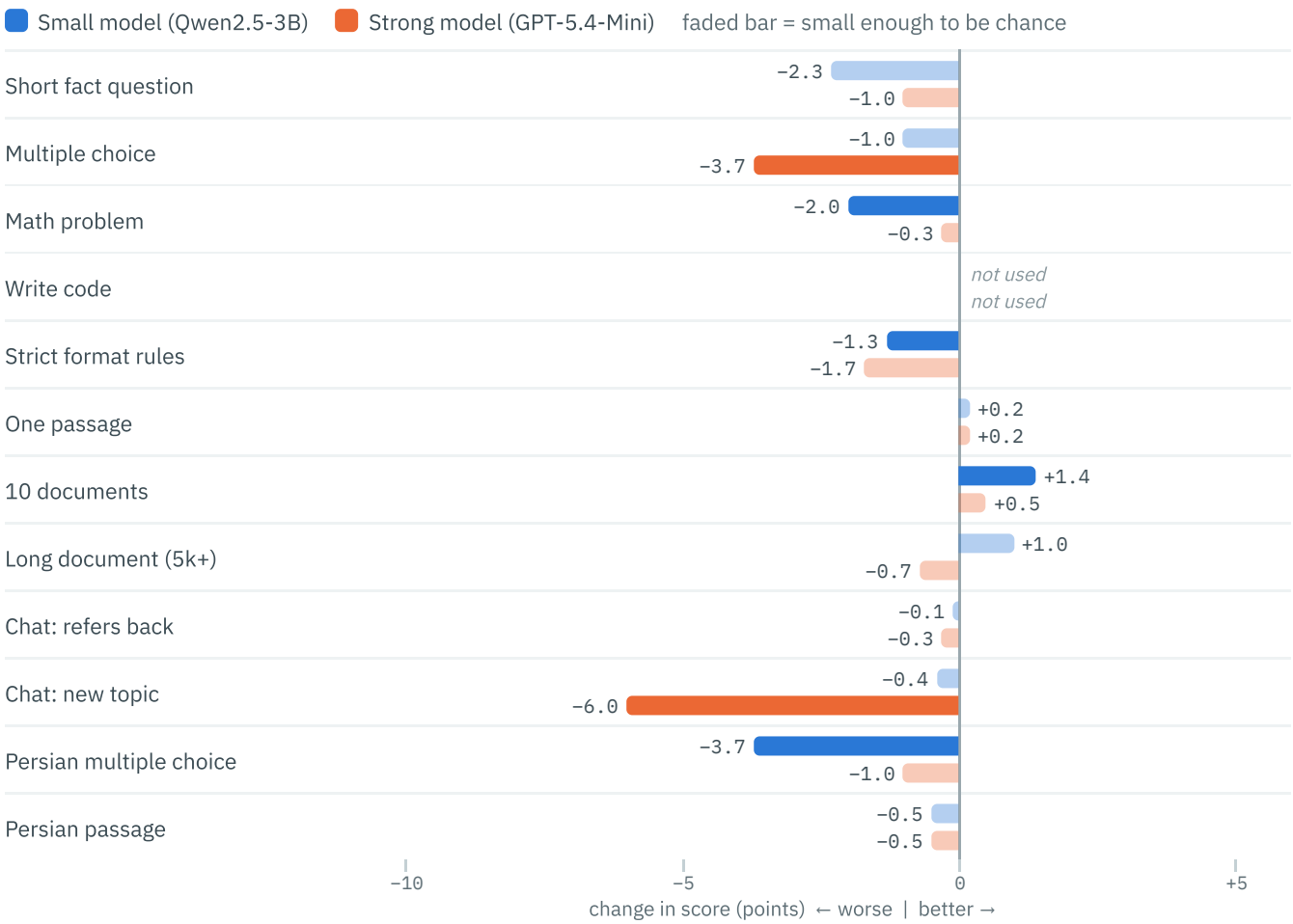
ANSWER WITHOUT METHOD	ANSWER WITH METHOD	CORRECT ANSWER
I don’t know.	Yuzuru Hanyu	Yuzuru Hanyu

With a long chat about Donald Duck in front of it, the model hesitated. After the chat was summarized, it answered.

5 · METHOD 3 · REWRITE THE QUESTION

A bigger rewriting model doesn’t fix it

The first test rewrote questions with our small local model (Qwen2.5-3B). A small model could be the real problem, so we ran it again with **GPT-5.4-Mini doing the rewriting** — same questions, same instruction, same safety check. Shipped settings:



What the strong model fixed. With the safety check off, the small model’s big losses (it even dropped “the comet comes every 75 years” from a math problem) were its own fault:

	small	strong
Strict format rules	−49.7	−9.7
Long document (5k+)	−23.0	−0.3
Multiple choice	−18.0	−3.3
Chat: refers back	−13.4	−2.0

What it didn’t fix. Rewriting still helped none of the 12 kinds of request. The questions in these datasets are already clear, so there is little to gain — and any change, even harmless polishing, can shift the answer (below). It also adds a full model call to every request.

Strong model, chat after a topic change: harmless polishing, answer lost

ORIGINAL QUESTION: `who played the dad in a christmas story`

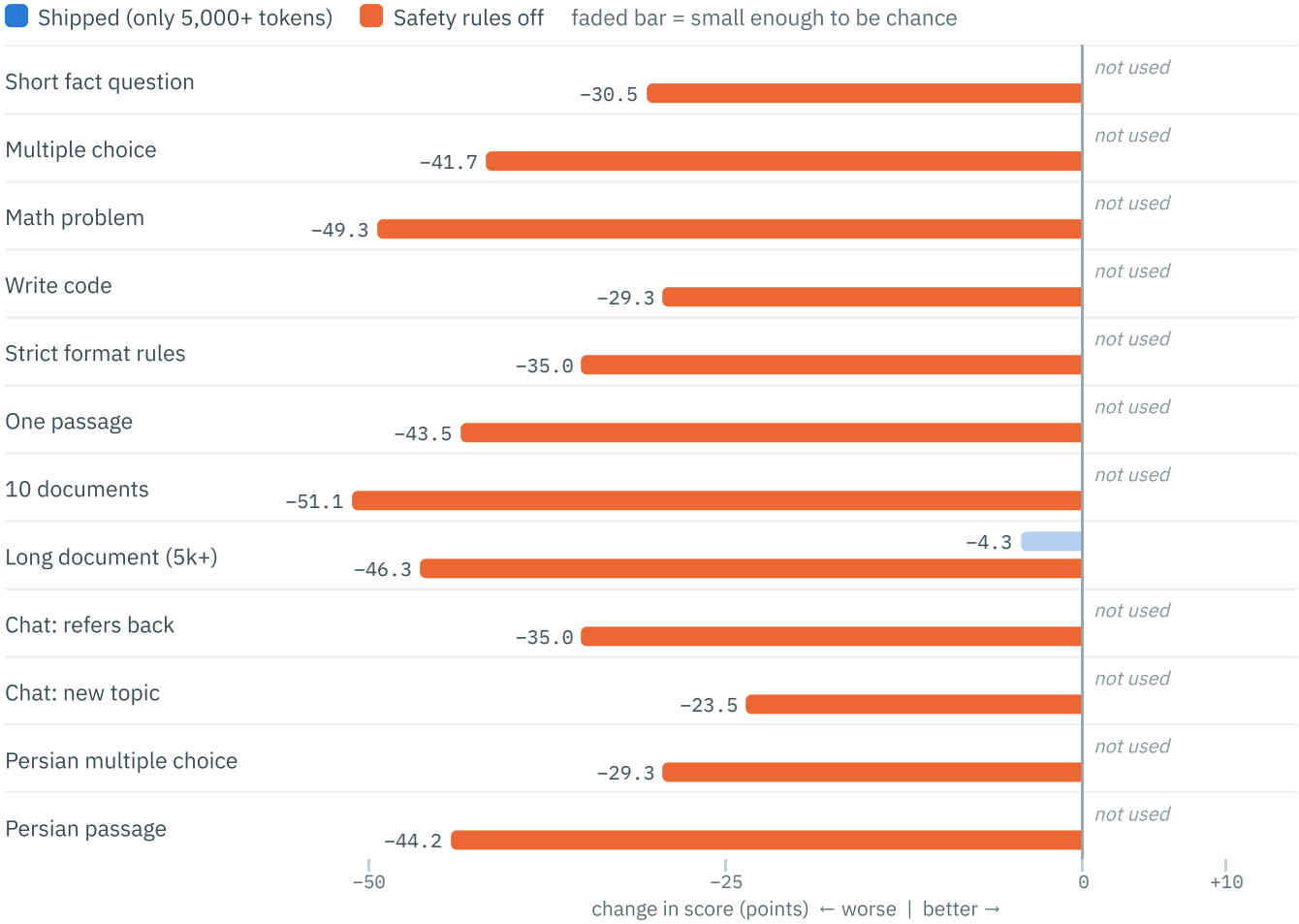
SENT AFTER THE METHOD: `Who played the dad in A Christmas Story?`

ANSWER WITHOUT METHOD	ANSWER WITH METHOD	CORRECT ANSWER
Darren McGavin	I don’t know.	Darren McGavin

Nothing was wrong with the rewrite. The answering model simply behaved differently — many of the lost answers here became “I don’t know”.

6 · METHOD 4 · REMOVE UNIMPORTANT WORDS

The 5,000-token rule is what keeps it safe



Short question, rules off: the question lost its meaning

ORIGINAL QUESTION: who wrote were going on a bear hunt

SENT AFTER THE METHOD: wrote bear hunt

ANSWER WITHOUT METHOD	ANSWER WITH METHOD	CORRECT ANSWER
Michael Rosen	Bear Hunt	Michael Rosen

On short text there is nothing unimportant to delete.

Persian, rules off: same problem

ORIGINAL QUESTION: در خانواده مجیدی به جز فرهاد چه کسی فوتبالیست بوده؟

SENT AFTER THE METHOD: خانواده مجیدی فرهاد فوتبالیست

ANSWER WITHOUT METHOD	ANSWER WITH METHOD	CORRECT ANSWER
فرزاد	مازندرانیِ قائم‌شهری	فرزاد

“Who else in the family, besides Farhad, was a footballer?” lost “who else” and “besides”.

7 · WHAT TO DO

Recommendations

1. **Turn on reordering.** Big cache gain on every kind of request, no quality loss anywhere.
2. **Keep summarizing off.** It saves 68% of chat tokens, but costs 43 points whenever the question depends on the chat. It would only be safe if the gateway could first detect that the user changed topic — that is future work.
3. **Keep rewriting off.** Even with a strong rewriting model (GPT-5.4-Mini) it improved none of the 12 kinds of request and made 2 worse (multiple choice −3.7, chat: new topic −6.0), while adding a model call to every request. Our small model was worse: it changed facts in math problems.
4. **Keep the 5,000-token rule for word removal, and don't lower it.** On long documents it halves the tokens for about 4 points of accuracy (the error margin reaches +0.3, so it may be less). Turn it on only where cost matters more than a small accuracy loss.

Bug found and fixed along the way. If the gateway policy merely *mentions* the personal-data placeholder format `<PII:TYPE:N>`, our safety check mistook it for a damaged placeholder and rejected every method on every request — all four methods would have silently switched themselves off in production. Fixed, with two new tests.

Limits of this test

- One answering model (GPT-5.4-Mini). A weaker model may be hurt more by the same changes.
- Dataset questions are already well written. Rewriting might do better on genuinely messy prompts; not tested.
- About 300 questions per kind of request, so changes smaller than ~3 points cannot be seen.
- Cache share is calculated exactly from the prompt text, not read from this model (it does not report cache use). An earlier live test on another model matched this kind of calculation (82% measured).
- Not tested: tool-calling agents, images, and very long chats (ours were a passage plus 12–20 short turns).

About this report. The text and layout of this report were written with the help of Claude (an AI assistant). The experiments themselves were really run: 3,464 requests per method on our test server, answered by GPT-5.4-Mini and graded against each dataset's own answer key. Every number here is generated directly from those results, and the raw results are saved alongside this report.